DT1 Kompletering

2 b)

$M = \{e, m, \{i, l\}, y, 0, x, s\}$

(i) No, because i is in a own set with l inside the set M.

(ii) Yes, because the set $\{i, l\}$ is a subset of the set M.

(iii) Yes, P(M) means all possible combinations of all the elements in the set. So $\{0, x\} \in P(M)$.

(iv) Yes, $\{\{i, l\}, 0, x\} \in P(M)$ because of the same explanation above.

(v) No because, $|M| = 7$, so, $|P(M)| = 2^7 = 128$. $256 \neq 128$.

8 a) $a_{n+1} = a_n + a_{n-1}$, for $n \geq 1$.

We know that $a_0 = 2$ and $a_1 = 1$.

When $n = 1$, $a_2 = a_1 + a_0 = 1 + 2 = 3$
When $n = 2$, $a_3 = a_2 + a_1 = 3 + 1 = 4$
When $n = 3$, $a_4 = a_3 + a_2 = 4 + 3 = 7$
When $n = 4$, $a_5 = a_4 + a_3 = 7 + 4 = 11$
When $n = 5$, $a_6 = a_5 + a_4 = 11 + 7 = 19$

$b_{n+1} = b_n + b_{n-1}$, for $n \geq 1$.

We know that $b_0 = 0$ and $b_1 = 1$

When $n = 1$, $b_2 = b_1 + b_0 = 1 + 0 = 1$
When $n = 2$, $b_3 = b_2 + b_1 = 1 + 1 = 2$
When $n = 3$, $b_4 = b_3 + b_2 = 2 + 1 = 3$
When $n = 4$, $b_5 = b_4 + b_3 = 3 + 2 = 5$
When $n = 5$, $b_6 = b_5 + b_4 = 5 + 3 = 8$

So,

$a_2 = 3$	$b_2 = 1$
$a_3 = 4$	$b_3 = 2$
$a_4 = 7$	$b_4 = 3$
$a_5 = 11$	$b_5 = 5$
$a_6 = 19$	$b_6 = 8$

8 b) Base case : when $n = 1$.

$a_1 - b_1 = 2b_0$, we know from the previous question that $a_1 = 1$, $b_1 = 1$, so it becomes $1 - 1 = 0$. we also know that $b_0 = 0$, so $2(0) = 0$. $0 = 0$.
Base case checks out.

Hypothesis: $n = k$.

$a_k - b_k = 2b_{k-1}$

Inductive step : $n = k+1$

$a_{k+1} - b_{k+1} = 2b_k$
we know from the previous question that $a_{k+1} = a_k + a_{k-1}$ and $b_{k+1} = b_k + b_{k-1}$ so we can rewrite it as:

$$\underbrace{(a_k + a_{k-1}) - (b_k + b_{k-1})}_{LHS} = \underbrace{2b_k}_{RHS}$$

We can rewrite LHS as: $(a_k - b_k) + (a_{k-1} - b_{k-1})$
now we apply hypothesis, so, $a_k - b_k = 2b_{k-1}$ and $a_{k-1} - b_{k-1} = 2b_{k-2}$, so, $2b_{k-1} + 2b_{k-2}$
$= 2(b_{k-1} + b_{k-2}) = 2b_k$ according to the reccurance formula for b.

Conclusion: $a_n - b_n = 2b_{n-1}$ for all $n \geq 1$ by induction.